

Heart Disease Dataset Features Explanation

Feature	Meaning	Values / Example
Age	Age of the patient in years	Example: 45
Sex	Gender of the patient	0 = Female, 1 = Male
Blood Pressure	Resting blood pressure (mm Hg)	Example: 120, 140
Cholesterol	Serum cholesterol level (mg/dl)	Example: 180, 220
Fasting Blood Sugar	Blood sugar level after fasting	0 = ≤ 120 mg/dl, 1 = > 120 mg/dl
ECG Result	Electrocardiographic results (heart electrical activity)	0 = Normal, 1 = Abnormal, 2 = Hypertrophy
Max Heart Rate	Maximum heart rate achieved during exercise	Example: 150
Exercise Angina	Chest pain during exercise	0 = No, 1 = Yes
ST Depression	ECG depression during exercise	Example: 0.0 – 6.0
Slope	Slope of ST segment in ECG	0 = Upsloping, 1 = Flat, 2 = Downsloping
Major Vessels (CA)	Number of major vessels visible in fluoroscopy	0 – 3
Thalassemia	Blood disorder affecting oxygen supply	0 = Normal, 1 = Fixed defect, 2 = Reversible defect, 3 = Severe

ANN Model Explanation

The **Artificial Neural Network (ANN)** model takes these health measurements as input. It analyzes patterns in the medical data and predicts whether the patient has a **risk of heart disease**.

The output is usually a **probability value between 0 and 1**:

- 0 → Low risk
- 1 → High risk

Higher probability values indicate a **greater likelihood of heart disease**.